

Assignment 3 – Numerical Integration and Differentiation

1. A dam is built across a river to make a reservoir in order to prevent floods and generate hydroelectric power. The depth of the reservoir (z) is given as a function f of x and y where $y = 0$ corresponds to the location of the dam (y is measured normal to the dam and x is measured along the dam). The file `reservoir.m` contains X , Y and the corresponding Z values on a 11×21 grid. To load this data, simply type 'reservoir' in your MATLAB script. Note that X is a 11×1 vector, Y is a 21×1 vector and Z is a 21×11 vector.

(a) Plot this data on a surface plot using the MATLAB function 'surf'.

(b) Compute the volume of the water V contained in the reservoir given by

$$V = \int_{-a/2}^{a/2} \int_0^b f(x, y) dy dx$$

In the data given $-a/2 < x < a/2$, $0 < y < b$ is a rectangular area with $a = 200$, $b = 400$. You can calculate this integral in **one** of three ways: (i) Simpson's rule, (ii) Matlab's **integral** function, (iii) trapezoidal rule for tabular data. *Hint:* When using either Simpson's rule or integral function, use the 'interp2' function (this should be used inside your function definition for f) to determine $f(x, y)$ for the sampled (x, y) points. To use the 'interp2' function, you must first use the 'meshgrid' function to project vectors X and Y to two 21×11 arrays that are the same size as Z .

(c) Compute the force F acting on the dam face given by

$$F = \int_{-a/2}^{a/2} \int_0^{f(x,0)} p(z) dz dx$$

where $p(z)$ is the pressure distribution on the dam face which depends only on z . $p(z)$ is given by

$$p(z) = \rho g z$$

Here, $\rho = 1000 \text{ kg/m}^3$ is the density of water, and $g = 9.81 \text{ m/s}^2$ is the gravitational acceleration. Substituting $p(z)$ in the force expression simplifies the double integral into a single integral given by

$$F = \frac{\rho g}{2} \int_{-a/2}^{a/2} f(x, 0)^2 dx$$

2. A three dimensional region is bounded by the following 6 planes: $x = 8$, $y = 2x$, $z = x + 2y$, $x = 0$, $y = 0$, $z = 0$ and the density of the region varies with the following function: $\rho(x, y, z) = \exp(-(x^2 + y^2 + z^2)/v(1 + z^3))$. Compute the mass M of this region given by

$$M = \int_{x=0}^8 \int_{y=0}^{2x} \int_{z=0}^{x+2y} e^{-(x^2+y^2+z^2)/\sqrt{1+z^3}} dz dy dx$$

using repeated applications of the Simpson's 1/3 rule with $n = 10$ segments in all directions. Compare your result with that from MATLAB's **integral3** function.

3. Develop a MATLAB function to integrate a function $f(x)$ in the interval $(-1,1)$ using n -point Gauss quadrature where $n = 1, 2, 3, 4, 5$. Use the attached Table to get the Gauss points and the corresponding weights. The formula for Gauss quadrature is given by

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i f(x_i)$$

For integrals with general limits from a to b you can use the following transformation:

$$x = \left(\frac{b-a}{2} \right) \xi + \left(\frac{b+a}{2} \right)$$

Use this function to compute the each of the following integrals using $n = 2, 3$, and 5 .

$$(i) \quad I_1 = \int_0^1 \sqrt{\ln\left(\frac{1}{x}\right)} = \frac{\sqrt{\pi}}{2}$$

$$(ii) \quad I_2 = \int_0^{\pi} x \ln(\sin x) = -\frac{1}{2} \pi^2 \ln 2$$

$$(iii) \quad I_3 = \int_0^1 \frac{1}{x(1-\ln x)\sqrt{\ln(1/x)}} = \pi$$

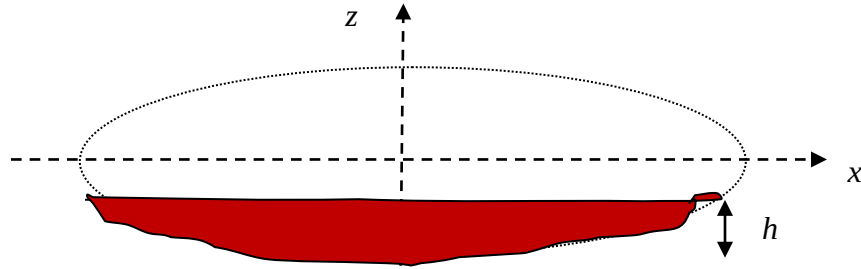
Compare the numerically computed values with the analytical solutions given above. Note that each integral has a singular point at one of the limits (*improper integral*) and hence a closed-type integration algorithm (e.g. Simpson's 1/3 rule) cannot be used.

4. Determine the volume of an ellipsoid described by the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

by using numerical integration using Simpson's rule with variable limits and check your results with the analytical value of $4\pi abc/3$. Use $a = 4$, $b = 2$, and $c = 1$ in your calculation.

A groundwater reservoir is contaminated with a dense non-aqueous phase liquid (DNAPL) having an approximate shape of a truncated ellipsoid with $a = 10$ m, $b = 4$ m, and $c = 2$ m. The height h of this shape is 0.9 m measured in the z direction from the bottom.



Calculate the mass of the DNAPL in Kg if its concentration C varies with z as follows:

$$C = \frac{C_0}{\sqrt{1+z^2}}$$

Assume $C_0 = 100$ mg/L in your calculation.

5. (i) Calculate the total contaminant mass in a cylindrical tank given a non-uniform concentration distribution:

$$c(r, z) = \frac{C_0 r^2}{\sqrt{z^2 + 1}}$$

Where r is the radius measured from the center and z is measured from the bottom. Tank radius (R) is 2 m and the fluid height is (h) is 3 m and $C_0 = 100$ mg/L.

(ii) Calculate the total mass in the tank after 1 hour if there is a volumetric influx of liquid into the tank at the following sinusoidal rate (m^3/s):

$$Q = Q_0 (1 + \sin(\omega t))$$

Assume the tank is empty in the beginning and the concentration distribution is the same as before and use $Q_0 = 100$ L/s, $\omega = \pi/6$, t is in seconds.

(iii) If the concentration is time dependent concentration in tank with the following distribution

$$c(r, t) = \frac{C_0 e^{-kt} r^2}{\sqrt{z^2 + 1}}$$

And the tank is empty in the beginning with $k = 0.001$, all other parameters remain same, (a) Plot total mass (M) vs time (t) for $t = 0$ to 3600 s; (b) Calculate the time averaged mass in the tank over this period.

6. The sediment oxygen demand [SOD in units of $\text{g}/(\text{m}^2\text{d})$] is an important parameter in determining the dissolved oxygen content of a natural water. It is measured by placing a

sediment core in a cylindrical container. After carefully introducing a layer of distilled oxygenated water above sediments, the container is covered to prevent gas transfer. A stirrer is used to mix the water gently, and an oxygen probe tracks how the water's oxygen concentration decreases over time. The *SOD* can then be computed as

$$SOD = -H \frac{dO}{dt}$$

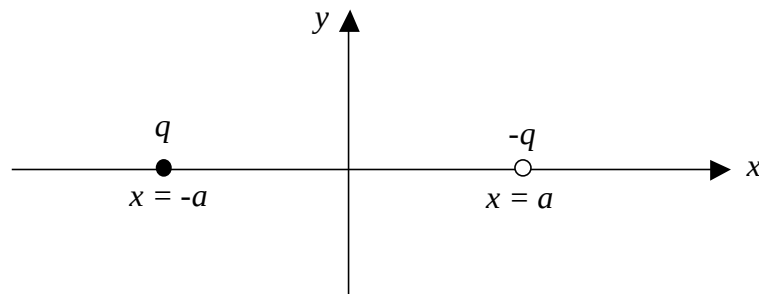
Where H = depth of water (m), O = oxygen concentration (g/m³), and t = time (d). Based on the following data and $H = 0.1$ m, use numerical differentiation (see examples provided on moodle, `jet_fighter.m` and `radar_tracking.m`) to generate plots of (a) *SOD* vs time and (b) *SOD* vs oxygen concentration.

t (d)	0	0.125	0.25	0.375	0.5	0.625	0.75
O (mg/L)	10	7.11	4.59	2.57	1.15	0.33	0.03

7. A well known groundwater flow problem is the re-circulating well problem where two wells, one recharge, and one pumping with equal rates, separated by a fixed distance are situated in a homogeneous infinite confined aquifer. The steady state head solution to this problem is given by

$$h(x, y) = h_{w0} + \frac{q}{4\pi K} \ln \frac{(x-a)^2 + y^2}{(x+a)^2 + y^2}$$

where $\pm q$ are the recharge and pumping rates, $\pm a$ is the x -coordinates (y -coordinates are assumed zero) of the wells, K is the hydraulic conductivity, and h_{w0} is the initial uniform hydraulic head in the aquifer before the wells are turned on.



In this problem, $a = 10$ m, $q = 100$ m³/day, $K = 1$ m/day, and $h_{w0} = 0$ m. Domain of interest is $x = y = -20$ m to 20 m. Plot the head contours for this field using the MATLAB's '**contour**' function. You may generate the mesh of x and y values in the domain of interest using the '**meshgrid**' function. Using Matlab's **gradient** function compute the velocity field given by Darcy's law:

$$v_x(x, y) = -K \frac{\partial h}{\partial x} \quad \text{and} \quad v_y(x, y) = -K \frac{\partial h}{\partial y}$$

Plot the velocity field in the same plot (head contour plot) using the '**quiver**' function. In a second plot, plot the contours of the potential function $\phi(x,y)$ and the stream function $\psi(x,y)$ given by

$$\phi(x,y) = \frac{q}{4\pi} \ln \frac{(x-a)^2 + y^2}{(x+a)^2 + y^2}$$

$$\psi(x,y) = \int \frac{\partial \phi}{\partial x} dy = \frac{q}{2\pi} \left[\tan^{-1} \frac{y}{x-a} - \tan^{-1} \frac{y}{x+a} \right]$$

These contour plots are known as equipotential lines and streamlines respectively. Also, note that the following relationship holds (you can check this in MATLAB if interested):

$$v_x = -\frac{\partial \phi}{\partial x} = -\frac{\partial \psi}{\partial y} = -K \frac{\partial h}{\partial x} \quad \text{and} \quad v_y = -\frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} = -K \frac{\partial h}{\partial y}$$

Number of Gauss Points	Location(s) of Gauss Points	Weight(s)
1	0	2
2	$\frac{1}{\sqrt{3}}$ $-\frac{1}{\sqrt{3}}$	1 1
3	0 $\sqrt{3/5}$ $-\sqrt{3/5}$	8/9 5/9 5/9
4	$\sqrt{\frac{3-2\sqrt{6/5}}{7}}$ $-\sqrt{\frac{3-2\sqrt{6/5}}{7}}$ $\sqrt{\frac{3+2\sqrt{6/5}}{7}}$ $-\sqrt{\frac{3+2\sqrt{6/5}}{7}}$	$\frac{1}{2} + \frac{1}{6\sqrt{6/5}}$ $\frac{1}{2} - \frac{1}{6\sqrt{6/5}}$ $\frac{1}{2} - \frac{1}{6\sqrt{6/5}}$ $\frac{1}{2} + \frac{1}{6\sqrt{6/5}}$
5	0 $\frac{1}{3}\sqrt{5-4\sqrt{5/14}}$ $-\frac{1}{3}\sqrt{5-4\sqrt{5/14}}$ $\frac{1}{3}\sqrt{5+4\sqrt{5/14}}$ $-\frac{1}{3}\sqrt{5+4\sqrt{5/14}}$	$\frac{128}{225}$ $\frac{161}{450} + \frac{13}{180\sqrt{5/14}}$ $\frac{161}{450} - \frac{13}{180\sqrt{5/14}}$ $\frac{161}{450} - \frac{13}{180\sqrt{5/14}}$ $\frac{161}{450} + \frac{13}{180\sqrt{5/14}}$